

Reading the H–R Diagram *by yourself*

An astronomer hands you a chart with thousands of dots and four axes. By the end of this lesson, you should be able to read it like a map.

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How to use this: Mark your own work against these model answers. Where your wording differs from mine, that's fine as long as the physics is right — these are *examples* of good answers, not the only correct ones. Watch-out notes flag common mistakes; if you made one, take a moment to figure out where your reasoning went wrong before moving on.

1 Working out *what the axes mean*

~ 8 MIN

1

NOTICE

Drag horizontally — what happens to the temperature?

MODEL ANSWER

The temperature gets **higher** as you drag *left*, and lower as you drag right. The hottest stars are on the left of the diagram. **This is the opposite of a normal graph**, where values increase to the right.

2

NOTICE

Drag vertically — by how much does L change at each gridline?

MODEL ANSWER

Each major gridline is a **$\times 10$ increase** in luminosity. The values go 0.001, 0.01, 0.1, 1, 10, 100, 1000... so the L axis is **logarithmic**, not linear. Crossing two gridlines means a $\times 100$ change.

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COMPARE

Record M at $L = 1, 100, 0.01$. What's the relationship?

MODEL ANSWER

$L (L_{\odot})$	M (ABSOLUTE MAGNITUDE)
1	+4.83
100	-0.17 (≈ 0)
0.01	+9.83

A **brighter star has a smaller (more negative) M** . The relationship is inverted: high luminosity $\rightarrow$ low magnitude. The scale is also *logarithmic* — each factor of 100 in L corresponds to 5 magnitudes ($M = 4.83 - 2.5 \log_{10}(L/L_{\odot})$).

WATCH OUT

It's tempting to write "magnitude = how big a star is". It doesn't — the word is unfortunate but historical. Magnitude in astronomy is purely a brightness scale.

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INSPECT

Spectral class order from hottest to coolest.

MODEL ANSWER

O B A F G K M (hottest to coolest). O-type stars are blue-white at $>30,000$ K; M-type stars are red at $\sim 3,000$ K. The Sun is class G.

PULL IT TOGETHER

The four conventions of an H - R diagram, in your own words.

MODEL ANSWER

1. **Temperature decreases left to right** (backwards). Hot blue stars on the left, cool red stars on the right.
2. **Luminosity is logarithmic**. Each major gridline is a factor of 10 in brightness.
3. **Absolute magnitude is inverted**: smaller numbers (more negative) mean brighter stars.
4. **Spectral class** labels the temperature axis with letters: O B A F G K M from hot to cool.

2 Mapping *where stars actually live*

~ 10 MIN

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OBSERVE

Where do most named stars sit? Cluster or scatter?

MODEL ANSWER

Stars **cluster** rather than scatter randomly. Most sit along a **diagonal band running from top-left (hot, luminous) to bottom-right (cool, dim)**. A few sit far above this band (very luminous), and a few sit far below it (very dim).

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INVESTIGATE

Record T , L , and a guess at what kind of star each is.

MODEL ANSWER

STAR	T (K)	L ($L_{\odot}$)	WHAT KIND
<i>Sun</i>	5,778	1.0	Main sequence (G-type)
<i>Sirius A</i>	9,940	25	Main sequence (A-type, hot)
<i>Betelgeuse</i>	3,500	126,000	Red supergiant
<i>Sirius B</i>	25,200	0.056	White dwarf
<i>Aldebaran</i>	3,910	440	Red giant

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CONFIRM

Did your guesses match? Why might any have been wrong?

MODEL ANSWER

The most common error is mixing up **red giants** and **red supergiants** — both are cool and red, so it's tempting to lump them together. The difference is luminosity: Aldebaran (red giant) is around $400 L_{\odot}$, while Betelgeuse (supergiant) is over $100,000 L_{\odot}$. *Supergiants sit higher.*

Another common error: thinking Sirius B is a main-sequence star because it's hot (25,000 K). But it's also dim ($0.056 L_{\odot}$) — that combination only makes sense for a *tiny* star, i.e. a white dwarf.

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DEFINE

One-sentence definition of each region.

MODEL ANSWER

- **Main sequence** — a diagonal band where stars are stably fusing hydrogen into helium; T and L are correlated (hotter stars are also more luminous).
- **Red giants** — cool ($\sim 3,000\text{--}5,000\text{ K}$), highly luminous ($10^2\text{--}10^4 L_\odot$) stars; large radii compensate for the cool surface.
- **Supergiants** — extremely luminous ($10^4\text{--}10^6 L_\odot$) across a wide range of temperatures; the largest stars known.
- **White dwarfs** — hot ($10,000\text{--}30,000\text{ K}$) but very dim ($\sim 10^{-2}\text{--}10^{-4} L_\odot$); their tiny radii ($\sim$ Earth-size) keep them faint despite the high temperature.

PULL IT TOGETHER

Most stars sit on the main sequence. Why? What does that tell you?

MODEL ANSWER

Most stars sit on the main sequence because **that's where they spend most of their lives** — the main sequence is the long, stable hydrogen-burning phase, which lasts billions of years for a Sun-like star. The other regions (red giants, supergiants, white dwarfs) are visited only briefly, or right at the end of a star's life.

So if you find a star on the main sequence, you can be confident it is currently **fusing hydrogen into helium in its core**, in stable equilibrium.

3 Discovering the link between *T, L, and R*

~ 10 MIN

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EXPERIMENT

Drag up at constant *T*. What happens?

MODEL ANSWER

The radius reading **increases** as you go up, and the star's drawn size visibly grows. So if temperature is fixed, more luminous stars must be **bigger**. This makes sense from $L = 4\pi R^2 \sigma T^4$: with *T* constant, $L \propto R^2$.

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EXPERIMENT

Drag horizontally at constant L . What happens?

MODEL ANSWER

The radius **decreases** as you move toward hotter temperatures (left). To keep luminosity the same while the surface gets much hotter, the star must *shrink* — because each square metre of a hot surface emits a lot more power ($\propto T^4$). From the equation: with L constant, $R^2 \propto 1/T^4$, so $R \propto 1/T^2$.

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REASON

If T and L both double, what happens to R ?

MODEL ANSWER

Rearranging $L = 4\pi R^2 \sigma T^4$:

$$R^2 = L / (4\pi\sigma T^4)$$

$$R^2 \propto L / T^4$$

If $L \rightarrow 2L$ and $T \rightarrow 2T$, then $T^4 \rightarrow 16T^4$. So $R^2 \rightarrow 2L / 16T^4 = (1/8) \times \text{original } R^2$. Therefore **$R \rightarrow R/\sqrt{8} \approx 0.35 R$** . The star is about **one-third its original radius**.

WATCH OUT

The instinct is "both go up so R goes up". But T is to the *fourth power* while L is only first-power, so doubling T overwhelms doubling L by a factor of 8. T dominates.

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APPLY

Use $L = 4\pi R^2 \sigma T^4$ to explain Rigel (hot, blue) vs Betelgeuse (cool, red), both $\sim 10^5 L_\odot$.

MODEL ANSWER

From the simulation: Rigel $T \approx 12,100$ K, $L \approx 120,000 L_\odot$. Betelgeuse $T \approx 3,500$ K, $L \approx 126,000 L_\odot$. Their luminosities are essentially the same.

For a fixed luminosity, $R^2 \propto 1/T^4$, so:

$$\begin{aligned} R_{\text{Bet}} / R_{\text{Rigel}} &= (T_{\text{Rigel}} / T_{\text{Bet}})^2 \\ &= (12100 / 3500)^2 \\ &= 3.46^2 \\ &\approx 12 \end{aligned}$$

Betelgeuse has a radius about 12× larger than Rigel's. Each square metre of Betelgeuse's cool surface emits much less power than each square metre of Rigel's hot surface, but Betelgeuse compensates by having a vastly greater surface area — about $12^2 = \sim 140\times$ more area.

PULL IT TOGETHER

Given just a star's position, what do you immediately know?

MODEL ANSWER

From a star's position alone you can read off:

- Its **surface temperature** T (from the x-axis)
- Its **luminosity** L (from the y-axis)
- Its **spectral class** (from the top axis)
- Its **absolute magnitude** M (from the right axis)
- Its **radius** R , calculated from L and T using $L = 4\pi R^2 \sigma T^4$
- Its **evolutionary stage** (which region it's in: main sequence, giant, supergiant, white dwarf)
- Roughly its **colour** (cool stars are red, hot stars are blue)

That's a *lot* of physics from one dot on a chart.

4 Following *the life of a Sun-like star*

~ 10 MIN

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TRACE

For each stage: position, what's happening, why it ends.

MODEL ANSWER

STAGE	POSITION	WHAT'S HAPPENING	WHY IT ENDS
Stellar Nebula	Top-right (cool, luminous)	Cloud of H gas + dust collapsing under gravity; not yet fusing	Gravitational PE → heat; core reaches 10^7 K and fusion ignites
Main Sequence	Middle of MS ($T=5778$, $L=1$)	H → He fusion in core; radiation pressure balances gravity (hydrostatic equilibrium)	Core hydrogen runs out; core contracts; helium fusion ignites
Red Giant	Top-right (cool, luminous)	He → C fusion in core; envelope expanded enormously; cool surface but huge area	Core helium exhausted; outer layers shed as nebula; hot core exposed
Planetary Nebula	Top-left (very hot, luminous)	Outer layers drift away as a glowing shell; tiny hot core revealed at the centre	Exposed core contracts under gravity until electron degeneracy halts it
White Dwarf	Bottom-left (hot, very dim)	No fusion; Earth-sized but Sun-massed; supported by electron degeneracy pressure	Doesn't end — slowly cools and fades over billions of years

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IDENTIFY THE CENTRAL IDEA

The recurring physical principle.

MODEL ANSWER

Hydrostatic equilibrium: a stable star is in balance between two opposing forces — **outward radiation pressure** from fusion in the core, and **inward gravitational pressure** from the star's own mass.

When fusion stops, radiation pressure drops, gravity wins, and the star contracts. When the contraction heats the core enough to start a new fusion reaction (e.g. helium fusion), pressure builds up again and a new equilibrium is reached. *Every transition in a star's life is a re-balancing.*

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REASON

Why can't a star stay on the main sequence forever?

MODEL ANSWER

Because **the fuel runs out**. Hydrogen in the core gets steadily converted to helium, and once the core hydrogen is exhausted, fusion stops. Without fusion, the core can no longer generate the radiation pressure that supports the star against gravity.

The equilibrium fails not because the gravity changes, but because *the outward push disappears when the fuel is gone*. The core then contracts, heats up, and the next stage of the star's life begins.

PULL IT TOGETHER – SKETCH

Sketch the path with each transition labelled.

MODEL ANSWER (DESCRIBED)

The path on the H–R diagram for a Sun-like star looks roughly like an "N" rotated:

1. **Nebula** (top-right) → contracts, heats up → joins **Main Sequence** (middle, where $T=5778$, $L=1$)
2. **Main Sequence** → core H exhausted, envelope expands and cools → climbs to **Red Giant** (top-right)
3. **Red Giant** → outer layers ejected, hot core exposed → moves *left* rapidly to **Planetary Nebula** (top-left)
4. **Planetary Nebula** → core contracts, no more fusion → drops down to **White Dwarf** (bottom-left)

Annotate transitions: "*H runs out* → *core collapses* → *He fusion starts*" between MS and Red Giant; "*He runs out* → *outer layers shed*" between Red Giant and PN; "*electron degeneracy halts collapse*" between PN and White Dwarf.

5 Mass *determines fate*

~ 10 MIN

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COMPARE

Compare the three masses across five characteristics.

MODEL ANSWER

	1 $M_{\odot}$	15 $M_{\odot}$	30 $M_{\odot}$
MS start	Middle of MS, G-type, $T \approx 5800$ K, $L=1$	Top of MS, O/B-type, $T \approx 30,000$ K, $L \approx 20,000$	Very top of MS, O-type, $T \approx 42,000$ K, $L \approx 200,000$
Lifetime	~ 10 Gyr (10 billion yrs)	~ 10 Myr (10 million yrs)	~ 5 Myr (5 million yrs)
Biggest stage	Red giant ($\sim 100 L_{\odot}$)	Red supergiant ($\sim 10^5 L_{\odot}$)	Red supergiant ($\sim 4 \times 10^5 L_{\odot}$)
How it dies	Sheds outer layers gently as planetary nebula	Core-collapse supernova	Core-collapse supernova
Remnant	White dwarf	Neutron star	Black hole

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SPOT THE RULE

Complete the four statements.

MODEL ANSWER

- (a) The more massive a star is, the **shorter** its main sequence life.
- (b) Stars with mass less than about **8 $M_{\odot}$** end their lives as **white dwarfs**.
- (c) Stars with mass roughly between **8** and **25 $M_{\odot}$** end as **neutron stars**.
- (d) Stars more massive than about **25 $M_{\odot}$** end as **black holes**.

WATCH OUT

The exact mass thresholds are approximate and depend on the star's metal content and the equations of state for nuclear matter. Different textbooks quote slightly different values (8 vs 10 $M_{\odot}$ for the WD/NS boundary; 20 vs 25 $M_{\odot}$ for NS/BH). The principle — that more mass means a more dramatic ending — is the key idea.

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PREDICT

Why do massive stars burn through fuel faster despite having more of it?

MODEL ANSWER

The chain of reasoning is:

1. **More mass** means stronger gravity pulling inwards.
2. To stay in equilibrium, the core has to **push back harder** — which means a higher core **pressure** and **temperature**.
3. Fusion rates rise **extremely steeply** with temperature (the proton-proton chain rate goes as T^4 ; the CNO cycle goes as T^{15-17}).
4. So a more massive star burns its hydrogen at a vastly higher rate — its luminosity is much higher, and its fuel runs out far faster than the extra mass can compensate for.

Concretely: a $15 M_{\odot}$ star has $15\times$ the fuel of the Sun, but burns $\sim 10,000\times$ faster — so it lasts only $\sim 1/1000$ as long.

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SYNTHESISE

A star at $T \approx 25,000 \text{ K}$, $L \approx 50,000 L_{\odot}$. Predict region, fate, and lifetime relative to the Sun.

MODEL ANSWER

(a) Region: This sits *just above* the main sequence at this temperature ($15 M_{\odot}$ MS is around $T=30,000$, $L=20,000$), and well below the supergiant region. It's likely a **blue supergiant** — a high-mass star that has just left the main sequence and is starting to evolve toward the red supergiant phase. (Accept also: *very high-mass main sequence star* — that's also defensible at this position.)

(b) Fate: A star of this luminosity (10^4 – $10^5 L_{\odot}$) corresponds to roughly **10–15 $M_{\odot}$** . It will eventually become a red supergiant, undergo a core-collapse supernova, and leave behind a **neutron star**.

(c) Lifetime: Massive stars at this part of the diagram have main-sequence lifetimes of around **10 million years** — about **1000× shorter than the Sun's 10 billion years**.

PULL IT TOGETHER

What makes the H–R diagram such a powerful tool? Why beat a table?

MODEL ANSWER

Two reasons:

- 1. It compresses many properties into a single position.** A point on the diagram simultaneously encodes T, L, R, M, spectral class, evolutionary stage, and (roughly) mass and lifetime. A table would need a column for each — and you'd have to read them off one by one.
- 2. It reveals patterns and stories that a table would hide.** The clustering of stars into regions, the diagonal of the main sequence, the evolutionary tracks across the diagram — these show that stars aren't a random collection of objects but follow predictable physical paths governed by mass. You see the *physics*, not just the data.